EV Charging VR Research Audit

Oct 2, 2026 · @Naman Khatak

A grounded audit of the EV-charging cybersecurity VR project (BeamNG.tech + Varjo XR-4 + Muse S Athena): what the codebase and data actually show versus what the mid-sem report and literature review claim, and a research plan built on that evidence.

Project understanding

The project (MTP-1, IIT Mandi, Arpita Kumari + Naman Khatak, supervisor Prof. Varun Dutt, mentor Ranik Goyal) studies how people make decisions at compromised EV charging kiosks inside a VR driving simulation, and whether behaviour, gaze and EEG can flag the risky ones. The participant drives an EV in BeamNG.tech with a 400-point budget, reaches a station of three kiosks, and can charge, skip, or ask an in-sim AI assistant for help. Some kiosks are malicious: a fake discount coupon, incomplete charge delivery, or overcharging. Eye movements are meant to be captured with a Varjo XR-4 (built-in eye tracking) and EEG with a 4-channel Muse S Athena.

The mentor's four original objectives were refined mid-semester into: (1) push simulation FPS up from ~40-50; (2) build a multimodal deep-learning pipeline over an existing 120+-participant dataset that drives AI warnings; (3) collect new data to validate the warning mechanism; (4) a literature review and gap analysis for this new approach. Scenario expansion (more attack types) moved to future scope, with the design work already done.

This audit reads differently from the mid-sem report in one respect: the report and its supporting literature review describe the plan and intentions well, but this audit also opened the actual BeamNG.tech installation, the Python backend, and the 140 session folders already on disk, to check what is really implemented and what the existing dataset really contains - not just what the documents say it contains. Several differences surfaced; they're called out explicitly in Objectives 1-3 below, each marked by what evidence supports it.

Objective status

Treating every objective as uncompleted unless the code or the data on disk shows otherwise.

Objective	Current status	Required work	Evidence
1. FPS optimization	Some application-layer polling found; engine source is closed (compiled Bin64 + asset-only gameengine.zip) - no baseline FPS measurement exists in the repo	Establish a reproducible baseline, profile CPU/GPU/physics split, apply config-level and IPC-cadence fixes	Verified: Bin64 is compiled .exe/.dll; game-settings-default.cs has live shadow/reflection prefs; receive_ui.py has two 5 Hz polling loops
2. AI pipeline + warnings	An image-based kiosk-cyberattack recommendation system (VLM screenshot analysis + RAG + LLM) is implemented and running - it already does what the report calls "AI help", just via visual anomaly detection rather than gaze/EEG. The automation-bias condition (malicious recommendation) is not yet built. EEG is now decoded for 67/140 participants (47.9%); gaze is complete for all 140	Add a malicious-recommendation condition to the existing assistant for the automation-bias study; fix the RAG store's cross-session contamination; build the gaze+behaviour baseline pipeline now that EEG has a real usable-n	Verified via ai_assistant.py, screenshot_analyzers.py, rag.py, ui_renderer.py, receive_ui.py; EEG verified via backend/Muse/decoded/decode_summary_resynced.csv (76/76 decoded, 67/76 usable)
3. New data collection	Not started (correct per report)	Define schema, synchronization and QC before collecting	Verified: no post-mid-sem session folders beyond the 140 already present
4. Literature review & gap analysis	Substantially complete and unusually rigorous - stronger than a typical mid-sem deliverable	Verify a handful of flagged citations (marked [to verify] in the report itself); extend to the corrected data picture	Verified by reading the review document in full - see Objective 4 below

The most consequential correction is inside Objective 2: the mid-sem report's Table 6 lists video, eye-tracking, EEG and event logs as available for all 120+ participants. The raw data on disk tells a different story - detailed in Objective 2.

Objective 1 - FPS optimization

The honest first finding: BeamNG.tech v0.38.3.0 is a closed commercial product here, not an open codebase. `Bin64/` holds compiled `.exe` / `.dll` files (BeamNG.tech.x64.exe, Vulkan layers, Ultralight), and `gameengine.zip` is assets only (cubemaps, datablocks, materials) - no C++ source. The render loop, physics stepper and VR compositor that the original brief asks for line-numbered fixes in are **not present in this install** and can't be profiled by reading code. What IS inspectable and genuinely "ours": the Lua gameplay

layer, the custom `EVChargingGame` BeamNG UI app, the Python backend that drives it via BeamNGpy, and the TorqueScript graphics preferences. Below is what that layer actually shows, each item graded by how confident the evidence is.

Architecture (confirmed by reading the code, not assumed):

Varjo XR-4 (VR display + eye tracker, not yet wired to the pipeline) -> BeamNG.tech engine (closed-source renderer + soft-body physics) -> `evProjectBridge.lua` GE extension (event-driven message bridge) -> `receive_ui.py` Python backend (BeamNGpy client, owns game state) -> `EVChargingGame` AngularJS UI app (`app.js` / `app.html`, runs in the CEF overlay) -> per-session CSV logs under `backend/raw_data/`.

Bottleneck candidates found, with exact locations:

Priority	File	Function / lines	What it does	Expected impact	Confidence
Quick win	`settings/game-settings-default.cs:31-32`	`$pref::Shadows::disable`, `$pref::Shadows::textureScalar`	Shadows run at full resolution/quality by default (disable=0, textureScalar=1.0)	Medium-High	Applied 2026-10-02 (textureScalar 1.0 to 0.5, disable 0 to 1); FPS effect not yet measured - no Windows/GPU access from this session
Quick win	`settings/game-settings-default.cs:22`	`$pref::Reflect::refractTexScale`	Reflection texture resolution was at full scale (refractTexScale=1.0) - frameLimitMS=0 was already the cheapest reflection setting, not uncapped; corrected from an earlier misread of that comment	Medium	Applied 2026-10-02 (refractTexScale 1.0 to 0.5); FPS effect not yet measured - no Windows/GPU access from this session
Quick win	`settings/game-settings-default.cs:38-42`	`$pref::BeamNGVehicle::dynamicReflection::*`	Dynamic vehicle reflections are already disabled by default - not a lever here	None (already off)	Verified
Medium	`backend/.../receive_ui.py:879-933` (`_ui_listener`)	`time.sleep(0.2)` loop issuing `queue_lua_command(..., response=True)` 5x/sec to read the UI bridge	Adds a synchronous Python<->Lua round trip every 200ms on a background thread	Low-Medium	Verified present; BeamNGpy's blocking-response semantics on the GE thread are UNVERIFIED without profiling
Medium	`backend/.../receive_ui.py:817-855` (`_odometer_worker`)	Same 5 Hz cadence, calls `vehicle.poll_sensors()` each tick	Second concurrent socket RPC stream at 5 Hz	Low-Medium	Verified present
Needs verification	VR compositor / Vulkan rendering, soft-body physics step	Not accessible (compiled binary)	Most likely the dominant cost given BeamNG's known architecture (CPU-bound soft-body physics + stereo VR rendering)	Very High	UNVERIFIED - requires running the profiler-equipped build on the actual Windows workstation

Priority	File	Function / lines	What it does	Expected impact	Confidence
Quick win	lua/ge/extensions/render/openxr.lua:164,174	openXRwindowViewMode (2D mirror view) / openXRdebugEnabled	Desktop mirror can composite both eye views; BeamNG's own UI labels "Between eyes" slowest vs "Empty" fastest. Separately, an OpenXR debug mode exists that explicitly reduces framerate - must be off before measuring.	Medium (mirror setting); any value (debug mode, if accidentally left on)	Developer's own code/UI labels, not yet measured - check current setting via ctrl+numpad0 before the Windows FPS run

The Varjo XR-4 panel refreshes at 90 Hz, so a stable 90 FPS is the correct, sufficient target - frames above the headset's refresh rate are never displayed. The report's stated 120 FPS target exceeds what the hardware can show and should be reframed as "stable 90, no drops," which is also the more defensible number to put in front of a reviewer.

Profiling methodology (since no baseline exists yet) - corrected. An earlier pass of this audit referenced a console command/overlay called `gfxFrameOverlay`; re-checked by searching the full codebase for that string and it does not exist anywhere in this install - that claim is retracted (logged as a self-correction). What's actually confirmed present by reading the code: `lua/ge/extensions/editor/perfProfiler.lua` is BeamNG's own native performance-recording viewer, reading `.perfRecording.json` files from `/perfRecordings/` with exactly fps_instant/average/min/max/waitForGPU, poly count, draw calls, and memory fields (confirmed from its own `dataKeys` table). Its companion recorder module, `test_util_fpsCamRecorder`, is referenced but its own file was not found in this install - UNVERIFIED how recording is actually triggered from gameplay; check BeamNG's developer documentation for this.
`lua/ge/extensions/render/openxr.lua` is BeamNG's OpenXR control layer and exposes a live debug window (toggle: ctrl+numpad0) reporting rendered-vs-recommended per-eye resolution and current-vs-target refresh rate - watching `currentRefreshRate` drop to half of `targetRefreshRate` is a direct, in-headset-visible signal that OpenXR reprojection has kicked in, which is a sharper diagnostic than a raw FPS number. Fall back to RenderDoc (GPU frame capture) or Windows Performance Analyzer (CPU) if neither native tool is usable. Any run should happen during a standard station-to-station drive, with and without the UI app and Python backend attached, to separate (a) base engine cost, (b) scene-content cost, and (c) IPC-layer cost - without that split, any specific "shadows vs. physics vs. IPC" claim beyond what's in the table below is a guess, not a finding, and the original brief's request for exact engine-level line numbers still can't be met from this install regardless of effort spent re-reading it.

Change applied, measurement still needed on your end. This session is macOS and `BeamNG.tech.exe` is a Windows binary with GPU dependencies, so the change below was made but not run or measured here - that has to happen on the Windows workstation:

`settings/game-settings-default.cs` now has `Shadows::textureScalar` 1.0 -> 0.5, `Shadows::disable` 0 -> 1 (Partial), and `Reflect::refractTexScale` 1.0 -> 0.5.

One caveat worth checking first: this file only sets defaults for a *fresh* preferences profile. If BeamNG already wrote a saved-preferences file for your Windows user account (outside this install folder, since `startup.ini`'s `UserPath` is blank/default), that saved file will override these defaults at runtime and the change won't show up until those specific settings are reset or re-applied through the in-game Options menu. Check your saved preferences before concluding the edit had no effect.

To measure: confirm OpenXR debug mode is off and the 2D mirror view is set to "Empty" (not "Between eyes", which BeamNG's own UI labels slowest - `lua/ge/extensions/render/openxr.lua:164,174`), then drive the same fixed route - spawn to first kiosk, same station count - twice: once on a git-stashed/reverted copy of the config file (baseline) and once with this edit (treatment), same build, same day, VR headset on both times since headset compositor load differs from a flat-monitor view. Record average FPS, 1% low FPS (via the native `.perfRecording.json` tooling if you can get it working, otherwise RenderDoc/Windows Performance Analyzer), and whether the shadow-quality drop is visually acceptable for a human-subjects study before locking it in - `Shadows::disable=1` (Partial) does change scene lighting, not just frame cost.

Objective 2 - AI / deep-learning pipeline

What is actually implemented today - corrected from an earlier pass of this audit (verified by reading `ai_assistant.py`, `screenshot_analyzers.py`, `rag.py`, `ui_renderer.py`, `receive_ui.py` in full, including the actual prompts, not just the dispatch logic). An earlier pass of this audit concluded the in-game AI assistant was "a screenshot+RAG deepfake-detection chatbot" unrelated to the report's described kiosk-trust warning system. That was wrong, and is retracted here. What it actually does: on each station, it renders the three kiosks to an HTML mock (including the coupon/discount banner image, drawn from the project's own `images/real/` vs `images/deepfake/` folders - "deepfake" here means a fake/AI-generated-looking *coupon graphic*, the project's own fake-discount-coupon attack, not an unrelated identity-deepfake task), screenshots it, and sends it to a VLM with the literal instruction: *"You are an expert in analyzing EV charging network user interfaces for signs of potential cyberattacks... Identify any*

anomalies or signs of cyberattack, such as deepfake or AI-generated images, unrealistic discounts, phishing prompts, or suspicious branding" (`screenshot_analyzers.py:11-36`). That structured per-kiosk anomaly analysis then feeds a "Help Agent" LLM prompt that is told to *"recommend the safest available kiosk for the user to use… Never recommend a kiosk that is occupied or unavailable"* (`ai_assistant.py:230-238`), optionally grounded with RAG retrieval over past charging events at the same kiosk image. This **is** the kiosk-safety recommendation system the mid-sem report describes - just image-based (VLM anomaly detection), not gaze/EEG-based, which is the one part of the earlier characterization that was correct: nothing in this code path touches gaze or EEG data. `aoi_mapper.py` exists (area-of-interest mapping groundwork) but nothing in `receive_ui.py` calls it yet.

What's actually missing for the automation-bias objective, now scoped precisely rather than "redesign from scratch": `_build_full_prompt` always instructs the LLM toward the *safest* recommendation - there is no condition variable for a deliberately malicious one. Adding a correct/absent/malicious condition flag (and, in the malicious arm, either suppressing the real anomaly signal fed to the prompt or overriding the final recommendation) is a scoped, bounded change to existing code, not new infrastructure.

A separate, previously undocumented risk found while tracing this: the RAG store (`rag.py`) persists to a single shared location (`backend/rag_index/`) across *every* session - `AIAssistant.reset_session()` (called at the start of each participant's run) clears in-memory state but never calls `RAGStore.clear()`. The index on disk currently holds 904 accumulated documents spanning dozens of sessions (confirmed by inspecting `rag_index/meta.json` directly), meaning a given participant's AI recommendations can be influenced by what happened in *other* participants' earlier sessions on the same (small, reused) set of coupon images - a real confound for any analysis treating participants as independent, unless `clear_rag_context.py` (which exists but is never called automatically) was in fact run between every session. Worth raising to the mentor now, not fixed unilaterally since it touches already-collected experimental data.

What the existing dataset actually contains (verified by inspecting all 140 session folders in `backend/raw_data/`, not just trusting the report's Table 6):

Modality	Report claims (Table 6)	What's actually on disk	Confidence
Behavioural / interaction logs	Available, 120+	Present for all 140 sessions (`<id>.csv` interaction log + `demographics.json` + `consent.json` + `system_config.json`)	Verified
Eye tracking (gaze)	Available, 120+	Present for all 140 sessions - the large per-session `<id>.csv` file (up to ~193k rows / 80+ MB) is actually Varjo gaze output (columns: gaze_forward_x/y/z, pupil diameter, eye openness, etc.), not a summary log as the filename suggests	Verified, and a genuinely good finding - gaze coverage is better than the report implies
EEG (Muse S Athena)	Available, 120+	**Not in `raw_data/` at all** - but found separately in a `backend/Muse/` upload, decoded, and session-linked: 67 of 140 participants (47.9%) have usable EEG. See the MUSE EEG data audit section below for the full, corrected trail - this row only describes `raw_data/` itself, where the report's Table 6 claim does not hold.	Verified - absent from raw_data/, present and decoded elsewhere (see MUSE EEG data audit)
Gameplay video	Available, 120+	**No video files found anywhere in raw_data.** The system instead captures single screenshots per station for the AI assistant	Verified absence
Data quality	Not yet assessed (per report, correctly)	One session is pre-labelled `session_..._126_glitched data` - a concrete, available example for the quality-check step	Verified

This changes Objective 2's scope: the "multimodal" pipeline currently has one reliable modality (gaze) plus behavioural logs, not three. EEG and video should be labelled **proposed for future collection**, not **existing data to analyze**, until the team locates EEG recordings or confirms none were kept with session IDs.

Data schema (reflecting what's verifiably available, not the idealized version):

```
Participant ID, Session folder, Trial/Station number, Kiosk chosen, Attack
type (Safe/Cyber Attack),
Attack mode (blocked/reduced), Discount coupon shown, Decision
(charge/skip/AI-help), Points lost,
Decision latency, AI-assistant used (bool), Gaze stream (raw_timestamp,
gaze_forward_xyz, pupil diameter,
eye_openness, status, stability) - present for 140/140, EEG stream
(TP9/AF7/AF8/TP10 @ 256Hz, +ACCGYRO/
OPTICS/BATTERY) - decoded and usable for 67/140 (47.9%), Outcome label
(risky/safe, from game logs only,
never from gaze or EEG)
```

Model plan, unchanged from the report's reasoning and still the right call given n~140: MLP baseline on fused features, then LSTM/GRU for the temporal decision sequence, a multimodal Transformer only if data quality supports it, each phase required to beat simple baselines (logistic regression, random forest, XGBoost) on participant-level held-out splits (GroupKFold / leave-participants-out). With EEG currently unverified as existing data, the first real pipeline iteration should be built and evaluated on gaze + behaviour alone, with EEG added later only if the team locates or newly collects synchronized recordings - this is a scope reduction, not a setback, and makes the n~140 overfitting risk the report already flags even more important to respect.

Evaluation: balanced accuracy / macro-F1 / AUROC (risky decisions are the minority class), an ablation across gaze-only vs. behaviour-only vs. combined, and participant-level cross-validation throughout - never a random trial-level split, which would let the model memorize individuals.

MUSE EEG data audit (update to Objective 2's EEG row)

New files landed in `backend/Muse/` - 13 zip archives. This updates, but does not replace, the earlier "zero EEG files found" finding: that was true of `raw_data/`, and is still true there; EEG was simply stored separately. The corrected picture, verified by opening the actual file contents rather than trusting filenames or folder names:

What's in the 13 zips:

- `G1.zip`, `G2.zip`, `G3.zip`, `Participants 1-10.zip` … `31-40.zip`: re-packaged duplicates of session folders already in `raw_data/` (behaviour + gaze only) - no new EEG content, just redundant exports.
- `Muse.zip`: raw EEG packet dumps for participants 008-040 (33 participants, no gaps).
- `Muse copy.zip` (superset of `Muse (1).zip`, which is a stale partial duplicate): raw EEG packet dumps for participants 041-088, **with a gap at 048-052** (5 participants have no file).
- `dataset (1-20).zip` / `dataset (21-40).zip`: the one well-organized set - per-participant folders that bundle the EEG `_muse.txt` file alongside that participant's gaze CSV, interactions CSV, demographics and consent, for participants 008-040. Participants 001-007 have no EEG file anywhere, consistent with EEG recording starting partway through data collection.
- `musedata.zip`: not experiment data - a developer's `musebandtests` tool repo (git history, Python venv, and its own bench-test recordings dated July 2026). Important anyway: it's what explains the raw format and proves a decoder exists (below).

Participant coverage: 76 of 140 raw_data sessions (54%) have an EEG file, by participant-ID filename match (IDs 008-047 and 053-088; missing 001-007 and 048-052; no EEG file found for any participant above 088, out of a dataset that now reaches at least participant 132).

File format, decoded from the actual bytes, not assumed: each `_muse.txt` is tab-separated `timestamp \t characteristic-UUID \t hex-payload` per line - a raw, undecoded Bluetooth LE notification dump, one line per packet. This exactly matches the output of `on_data_notify()` in `musebandtests/collect_data.py`, which pulls data via the `OpenMuse` library (`OpenMuse.decode.parse_message`) and `bleak`. That same repo has already-decoded output from its own bench tests (`eeg_decoded.csv`, `eeg.csv`) proving the format is decodable with existing, working code:

Property	Value	How verified
EEG channels	TP9, AF7, AF8, TP10 (+ AUX in some decoded samples)	Header row of `eeg_decoded.csv` / `eeg.csv` in the musebandtests repo's own test recordings

Property	Value	How verified
Sampling rate	~256 Hz	Computed directly: a 60.13s musebandtests test recording logged 15,422 EEG samples (15422 / 60.13 = 256.5 Hz) - not taken from a spec sheet
Other available streams	6-channel motion (accel/gyro), 16-channel optics (PPG/fNIRS)	Confirmed present in all 76 production files once actually decoded (see Decode results below) - an earlier pass of this audit wrongly concluded only EEG was captured, based on seeing a single BLE characteristic UUID; the sensor-type distinction turned out to be inside the packet payload, not the UUID. Retracted.
Decoder availability	Working (`OpenMuse` package + `process_and_playback.py`)	Confirmed present in `musedata.zip`, with demonstrated raw-to-decoded output in the repo's own recordings
Signal quality indicators	None yet computed	The raw hex format carries no quality metric; one is only obtainable after decoding - not done yet

Cross-modality check - spot-checked 2 of 76, not all: participant 021's `_muse.txt` runs 2026-04-23T10:42:39 to 10:44:41 UTC (16:12:39-16:14:41 IST), which falls inside `raw_data/session_20260423_160401_021`'s own logged window (16:06:57 onward) - genuine wall-clock overlap between two independently-timestamped systems, not a filename coincidence. Participant 041 has three EEG file parts (`041` , `041_2` , `041_3`) - a 1.4-second failed connection, then a ~10-minute segment, a ~3-minute gap, then another ~11-minute segment - all falling within session `session_20260622_155420_041`'s window. That gap is a real, observed BLE dropout/reconnect, not an artifact of the audit. Both spot checks pass; the other 74 EEG-bearing participants have not been individually checked yet.

Verdict on Behaviour + Gaze + EEG as a triad, stated precisely per the task's categories:

Modality combination	Status
Behaviour + Gaze, all 140 sessions	Verified
Behaviour + Gaze + EEG(raw), for the 76 participants with an EEG file	**Verified and usable for 67/76 (47.9% of the full 140-session dataset)** - all 76 decoded; session-overlap confirmed for 69/76 after fixing a duplicate-session-folder bug in the cross-check (see Decision D019); applying an inclusion rule (overlap confirmed, excluding 2 dropout-heavy files and 1 zero-sample file, flatline fraction under 15%) lands at 67 usable participants. Sample rate (256.0 Hz) and full channel schema (EEG + motion + PPG/fNIRS + battery) confirmed directly from the production files. Per-sample synchronization against gaze/behaviour event markers is still not done - see Decode results below.
Behaviour + Gaze + EEG(raw), for participants 001-007, 048-052, 089 and above	Unavailable
Whether `collect_data.py` /OpenMuse (not `muse.py`) was the actual tool used to record the 76 participant files	Uncertain - the raw-packet line format is an exact match, which is strong circumstantial evidence, but no session-ID or participant-ID field is embedded inside any `_muse.txt` file itself to confirm it directly

This is a genuine, material upgrade from the prior "EEG absent" finding - but "a decodable raw file probably exists for just over half the participants" is a different claim from "we have a multimodal dataset," and the two should not be conflated in the report or the manuscript.

Decode results (update - all 76 files actually decoded)

Ran `OpenMuse.decode` against all 76 participants' raw files. One real bug surfaced and was worked around: `decode_rawdata` crashed on 23 of 76 participants with a numpy broadcast error, traced to a handful of malformed OPTICS subpackets (truncated BLE packets, wrong channel count - 30 bad subpackets out of roughly 27 million EEG samples processed, i.e. about 0.0001% of the data) that aborted the *entire* decode, EEG included, not just OPTICS. Fixed by decoding each sensor stream independently and dropping only the malformed subpackets within it. After the fix: **76/76 decoded successfully.**

- **Sample rate confirmed exactly: 256.0 Hz** (median and mode across all 76 - matches the Muse spec, now measured rather than estimated).
- **All four sensor streams are present and decodable**, not just EEG as first thought: EEG (76/76), ACCGYRO - 6-axis motion (76/76), OPTICS - 16-channel PPG/fNIRS (76/76), BATTERY (75/76). This corrects the earlier audit claim that "only the EEG characteristic was captured" - all streams were multiplexed inside that one BLE characteristic and only showed up on actually decoding the packets, not from the outer UUID.
- **Session-overlap cross-check extended from 2 participants to all 76, then corrected**: the first full pass found 67/76 overlapping, with 6 mismatches (012, 053, 054, 055, 058, 082). Investigating those 6 found a real bug in the cross-check script itself: 2 participants (055, 082) have *more than one* `session_*_<id>` folder on disk (082's first folder is a 9-row near-empty log; its real, 445-row session is a second folder 20 minutes later that the script wasn't checking), and once every candidate folder is checked, both resolve cleanly - **corrected total: 69/76 confirmed overlap**. Of the remaining 4: three (012, 053, 054) show a short EEG recording ending just minutes *before* their session's logged start - consistent with a pre-session headset check that was never followed by real in-session recording. One (058) has only a single, 2-line, essentially empty interactions log on disk - likely a logging gap (the real played session isn't captured under this participant ID anywhere found), not necessarily bad EEG. 2 more (059, 072) have no readable interactions CSV to compare against at all. 1 (013) decoded zero EEG samples despite 21,119 raw lines - likely a failed/empty connection.
- **Recording duration**: median ~23 minutes, IQR 12.7-33 minutes. Two participants (072, 084) show a nominal duration inflated by one enormous single gap each (072: ~3.6h nominal, ~3.2h of that is one gap, ~26 real minutes of data; 084: ~74min nominal, ~49min of that is one gap, ~24 real minutes of data) - these should not be treated as long clean recordings without accounting for the gap.

- **Signal quality (flatline fraction, a proxy for a dead/disconnected channel)**: median 1.0% of samples flat per channel, max 16.8% (participant 065) - no channel is anywhere near fully flat; nothing here disqualifies a participant outright, but 065, 053 and 037 are worth a closer look before inclusion.
- Decoded CSVs (`<id>_eeg_decoded.csv`, `_accgyro_`, `_optics_`, `_battery_`, plus `decode_summary.csv` with every metric above per participant) are written to `backend/Muse/decoded/` (7.7 GB, not committed to the doc).

Revised verdict, with the number now computed: a real, decoded, participant-linked EEG (+ motion + PPG) dataset exists for a working subset of participants. Applying an inclusion rule - session-overlap confirmed, excluding the two dropout-heavy files (072, 084, where most of the nominal duration is one giant gap) and the one zero-sample file (013), flatline fraction under 15% - gives **67 of 140 participants (47.9%)** with usable EEG. That threshold is a reasonable, stated judgment call (the median flatline fraction is 1.0%, so a 15% cutoff is generous, not arbitrarily tight on otherwise-good data), not an independently validated one - a different threshold would move the number, and that should be acknowledged if challenged. See Decisions D013, D019.

Gaze pipeline pilot (fixation/saccade detection)

Gaze is complete for all 140 sessions and isn't blocked by anything above, so this was piloted on 3 participants (021, 041, 100) before committing to a full run. Method: I-VT (velocity-threshold) classification on the Varjo `gaze_forward` vector at its native ~199-200 Hz (empirically confirmed from the data's own timestamps), filtered to Varjo's own `status==2` and `stability>0` fields.

Participant	Valid sample fraction	Fixations	Median fixation duration	Mean saccade amplitude
021	63.1%	172 (103/min)	158 ms	2.14 deg
041	64.9%	2243 (82/min)	195 ms	4.17 deg
100	61.7%	1122 (79/min)	170 ms	1.99 deg

All three land inside normal physiological ranges reported in the eye-tracking literature (fixation durations ~150-250ms; a right-skewed saccade-amplitude distribution) - the pilot's sanity check passes. The ~35-39% invalid-sample rate is real and quantified (Varjo's own quality fields, mostly `stability==0.0`), not a detection bug, and needs a decision before scaling (exclude those windows, or use too-high invalid-fraction as a session-level exclusion criterion).

AOI mapping (which kiosk, the coupon, or the price someone is looking at) is structurally blocked, not just unbuilt. The gaze CSV has only *direction*, not scene geometry - there is no synchronized video or logged 3D/screen-space position of the kiosk UI elements anywhere in this dataset to ground a screen-space or world-space AOI assignment. This is the literature review's own flagged top technical risk ("dynamic AOI mapping in VR"), now confirmed as a real, specific blocker rather than a generic caveat. It needs either kiosk screen-space coordinates logged going forward (a code change to the flowgraph/BeamNGpy layer), or manual region calibration against representative screenshots for the existing 140 sessions - a decision for the team, not something this pipeline can route around on its own. See Decision D020.

Objective 3 - data collection requirements

You're handling the physical collection yourselves, so this is deliberately just the spec the next round needs to produce scientifically usable data - not a logistics plan.

- **Correction: `muse.py` is not the tool that produced the existing EEG data.** The 76 participant `_muse.txt` files in `backend/Muse/` are raw BLE packet dumps matching `musebandtests/collect_data.py`'s format (`OpenMuse` + `bleak`, direct Bluetooth) - `muse.py` uses `pylsl`, a different capture path never run against these files. Wiring `muse.py` in as planned would create a second, incompatible EEG format rather than continuing the one that already decodes. Recommendation: adapt the `collect_data.py` / `OpenMuse` approach instead, with session-ID-tagged output on the same clock as the gaze/behaviour logs - see Decision D009.
- **Synchronization**: a shared timestamp or event marker across the gaze stream, the EEG stream, and the `interactions_*.csv` log, anchored on identifiable events already logged (`Flowgraph Reached`, `StationGenerated`, charging start/end) - `receive_ui.py` already timestamps every incoming/outgoing message (`_log_interaction`), so the event-marker backbone exists; it just needs the EEG stream joined to it.

- **Event markers already available in the logs**: `Started`, `Reached` (station N), `AudioTrigger`, `DrivingPageReached`, `ChargeUpdate`, `OutOfFuel`, `Finished` - these are the natural segmentation points for "approach -> notice -> decision -> response" windows.
- **Minimum required per session**: consent + demographics (already standard), gaze CSV, EEG CSV (once wired), interaction CSV, a system_config snapshot (already standard), and a pass/fail quality flag per modality rather than a single session-level pass/fail - the `126_glitched data` session shows a session can be fine on some modalities and bad on others.
- **Quality-control checks before a session counts toward the usable n**: gaze tracking-status column not predominantly invalid, EEG contact quality above threshold, event-marker count matching expected station count (should be ~20 `Reached` events per session), no duplicate/corrupted session folders (the repo already has at least one: `session_20260428_140404_032` has both `032.csv` and `032_1.csv`).
- **Clear the RAG store between sessions.** `rag.py`'s `RAGStore` persists to a single shared `backend/rag_index/` across every participant, and `AIAssistant.reset_session()` never calls `RAGStore.clear()` - confirmed on disk as 904 accumulated documents spanning dozens of sessions. Any new collection round should run `clear_rag_context.py` before each session, or the fix should be made automatic in `reset_session()`.
- **Log kiosk screen-space or world-space coordinates per frame** if AOI-level gaze analysis (dwell on coupon vs. price vs. kiosk screen) is wanted. The existing gaze stream has direction only - nothing in this dataset grounds it to a specific UI element. This is a code change to the flowgraph/BeamNGpy layer, not a data-analysis task.

Objective 4 - literature review synthesis

The literature review document already provided ("Literature Review & Research-Gap Analysis") is unusually rigorous for this stage - sharper and more self-critical than most mid-sem reviews, and this audit builds on it rather than re-running the search. It covers seven domains (EV cybersecurity, human factors, eye tracking, EEG, EEG+eye fusion, VR/driving-simulator cybersecurity, AI trust/automation bias) with a structured evidence matrix (paper, participants, modality, finding, limitation) rather than a loose bibliography, and it distinguishes peer-reviewed sources (Nasr et al. 2022, Thomopoulos et al. 2024, McAlaney & Hills 2020, Horowitz & Kahn 2024) from items it itself flags `[to verify]` (the

2023 driving-sim ransomware/trust study, Soykan et al. 2021, Kurt et al. 2026's final DOI, the LSL citation) - that honesty about unverified citations is exactly the standard the original brief asks for, already applied.

Its most load-bearing, defensible claims: EV-charging cybersecurity research is almost entirely technical (Nasr et al.'s 16-system pentest, zero human subjects); phishing eye-tracking/EEG work is real but confined to static desktop screens; EEG+eye-tracking fusion research exists but is small and fragmented (Thomopoulos et al. screened 651 papers down to 18); and the single closest prior work, Kurt et al. (2026, maritime GNSS spoofing, 61 seafarers, EEG+eye-tracking), is a methodological twin but in a different domain, with passive detection rather than an active kiosk choice.

Two things this audit adds on top of it, both from the codebase/data check above:

- The review's EEG caution ("4-channel Muse, treat as exploratory/secondary") turns out to be more severe in practice than in theory - the current dataset has no EEG files at all, so the fusion literature it cites (gaze alone often matching or beating EEG+gaze fusion) is not just a risk to manage, it's close to the default outcome unless EEG gets collected and wired in.
- The review correctly privileges gaze as the primary modality; the data check confirms gaze coverage is actually better than the report elsewhere implies (all 140 sessions, not a small pilot), which strengthens rather than weakens the review's own recommended framing.

No new domain search is warranted at this stage - the gap is well-evidenced. What's worth doing next is verifying the four `[to verify]` citations before they appear in a manuscript. One correction to an earlier pass of this audit: the in-sim AI assistant is not a side-topic closer to "phishing/deepfake-detection UX research" - it already is a kiosk-cyberattack-detection-and-recommendation system (VLM anomaly analysis -> safety recommendation), image-based rather than gaze/EEG-based, and belongs squarely inside the review's own decision-making frame, not folded in as an adjacent topic.

Research gap

Carried over from the existing review (it is already evidence-based, not a manufactured "few studies combine X and Y" claim), with the data-reality check folded in.

Area	Existing research	Weakness of prior work	Opportunity here
EV cybersecurity	Extensive, technical (Nasr et al. pentested 16 real systems)	No human subjects anywhere in the technical literature	First human-in-the-loop EV-charging deception study
Human factors in security	Extensive	Almost entirely email/static-screen phishing	Interactive HMI deception + genuine EV charge-anxiety urgency
Eye tracking + security	Moderate (phishing)	Attention != accuracy; static AOIs only	Dynamic 3D VR AOIs, decision-linked gaze - and this project actually has gaze data for all 140 sessions to work with
EEG + security	Moderate	Needs many channels/trials for reliable signal	Feasible only as exploratory/secondary on a 4-channel Muse - and not yet collected here, which the research plan should say plainly
EEG + eye-tracking fusion	Small, fragmented (18 papers per Thomopoulos et al.)	Lab-static settings, small n	Answers the field's own call to extend into a realistic multi-sensory environment - once EEG exists
VR + cybersecurity	Exists	Almost entirely education/training, not measurement	A measurement paradigm instead of a training one
EV + human factors	Essentially absent	-	The core, well-evidenced gap

Area	Existing research	Weakness of prior work	Opportunity here
AI trust / automation bias	Robust behaviourally (Horowitz & Kahn, 9,000 participants)	No physiological measurement toward a *malicious* recommendation, anywhere, let alone in EV charging	A clean, cheap, within-subject manipulation (correct / absent / malicious AI advice) this project's existing assistant architecture can support with modest changes

The gap is genuine and well-supported. The risk is not the gap - it's claiming more coverage of it (EEG, video) than the current dataset actually delivers.

Novelty analysis

Candidate claim	Closest existing work	What's missing from it	Genuinely novel here?
"First to combine VR + EEG + eye tracking + AI"	Several partial combinations exist	N/A - this is a tool-count claim, not a contribution	No - avoid this framing entirely, as the review already insists
Human-in-the-loop study of interactive EV-charging deception, measured behaviourally + via gaze	Kurt et al. 2026 (maritime GNSS spoofing, EEG+eye, 61 seafarers, *passive detection*)	Different domain; passive detection, not an active multi-option choice	Yes - domain + the active-choice paradigm are both real differences

Candidate claim	Closest existing work	What's missing from it	Genuinely novel here?
...with EEG as a confirmed modality	Kurt et al. 2026, Thomopoulos et al.'s 18-paper fusion corpus	Both have real EEG data; this project currently does not	Closer, not yet fully defensible - EEG is now decoded and session-linked for 67/140 participants (47.9%), a real dataset rather than an aspiration, but still a secondary-modality subset of the full sample, not full-cohort multimodal coverage. Claiming parity with Kurt et al.'s dedicated EEG+eye study would still overstate it.
Automation bias toward a malicious (not just absent/correct) AI recommendation, with physiology	Horowitz & Kahn 2024 (behavioural, 9,000 participants, no physiology, not EV)	No physiological measurement of trust toward a malicious automated recommendation, in any domain found	Yes, and cheap to add - the existing AI-assistant architecture (Objective 2) is one condition-variable away from supporting correct / absent / malicious recommendation arms
VLM-based kiosk-cyberattack recommendation (corrected from "deepfake detection" - it IS the kiosk-trust system, see Objective 2)	Not what any of the reviewed EV-security or phishing-eye-tracking literature studies	Not a different research question from the report's own - a different *modality* for the same kiosk-safety-recommendation idea (image-based, not gaze/EEG-based)	The automation-bias extension (correct/absent/malicious recommendation, see above) is the novel piece; the VLM-recommendation system itself is useful existing infrastructure for it, not a separate novelty claim

The defensible core claim stays what the existing review already settled on: a human-in-the-loop investigation of decision-making under interactive EV-charging deception, in immersive VR, measured through behaviour and gaze (EEG and the automation-bias sub-study as stated extensions, not assumed ones).

Proposed research direction

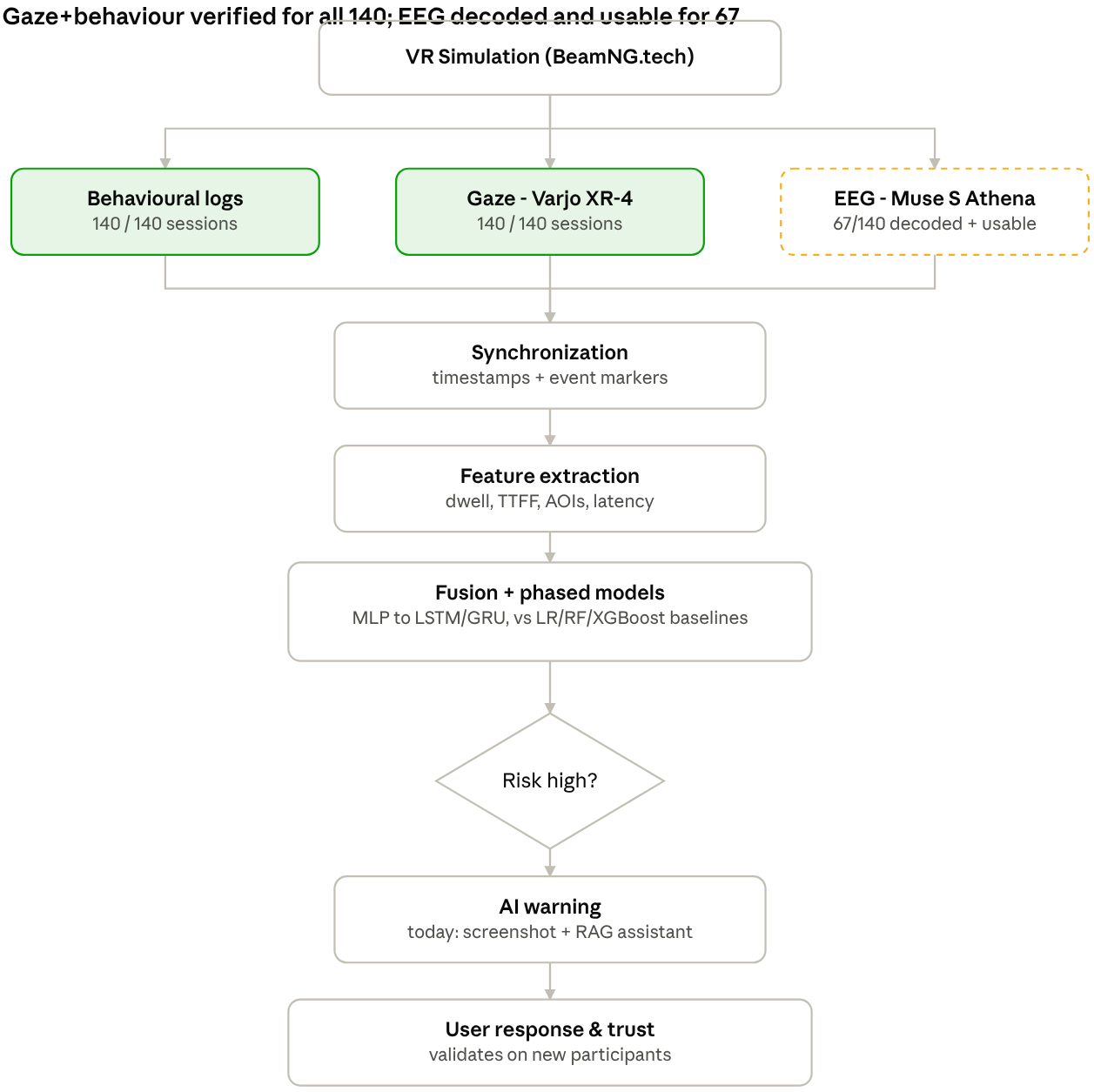
Primary research question: Does the presence and type of a simulated EV-charging cyberattack change decision outcomes and gaze-based attention patterns, and can those patterns, evaluated on held-out participants, flag risky decisions?

This is the review's own reframing of the report's un-testable "how can signals be used to identify vulnerability" question into one with a manipulable independent variable (attack present/absent, attack type) and measurable dependent variables (decision, latency, points lost, gaze dwell/TTFF on the coupon and price AOIs). It is chosen over the EEG-centred framing because it is the question the *existing* dataset (140 sessions, gaze + behaviour verified present) can actually answer today, without waiting on new EEG collection.

Secondary question (automation bias): Does a malicious AI-assistant recommendation increase unsafe kiosk choices relative to no recommendation or a correct one, and does independent verification behaviour (gaze to alternative kiosks/cues) mediate that effect? This reuses the existing AI-assistant code path (Objective 2) with one added condition variable, making it the cheapest genuinely novel extension available.

Phased, not deferred - EEG now has a real starting n. The gaze+behaviour paper is still the first, primary analysis (full 140-participant coverage, no open data questions). A second, EEG-extended analysis can proceed in parallel on the 67 participants (47.9%) with decoded, session-linked, quality-screened EEG - a real dataset, not a future promise, though still a subset rather than full-cohort coverage, and per-sample (not just session-level) synchronization against gaze/behaviour event markers is still outstanding. For any *new* data collection, EEG capture should use the `collect_data.py` / `OpenMuse` approach (confirmed to be what actually produced the existing 67-participant dataset), not `muse.py`, which uses a different, unused capture path (LSL) - see Decision D009. Stating this as two phases (gaze+behaviour paper first, EEG-extended analysis alongside it on the existing subset, new EEG collection after that) is more defensible to a reviewer than promising a three-modality study on the full sample the data doesn't support.

Complete research architecture



pipeline architecture - gaze and behaviour verified present for all 140 sessions, EEG not yet wired

The low-risk branch (not drawn) loops silently back to the next kiosk with no warning shown; only the high-risk branch reaches the AI-warning box. "User response & trust" at the bottom is the new-data-collection validation step, not a live feedback loop into the current session.

Experimental plan

Design: within-subjects - each participant already sees a mix of safe and attack kiosks across ~20 stations per session, so the existing 140-session dataset already has the structure a within-subject analysis needs; the automation-bias sub-study (correct / absent / malicious AI advice) should also be within-subject, randomized and counterbalanced across stations.

Sample size reasoning: n=140 (verified on disk) is large by the standards of comparable gaze/EEG human-factors studies (Kurt et al.: 61; VR driving-EEG studies: 15-31; phishing eye-tracking studies: 22-70) - more than adequate for mixed-effects behavioural/gaze analysis, but not large enough to justify a deep multimodal model as the primary analysis. EEG is decoded and usable for 67 of the 140 participants (47.9%) - a real, quality-screened subset, not a future promise - so any EEG-involving analysis should budget for n=67, not 140, and should still be treated as a secondary track. Deep learning (MLP/LSTM/Transformer) should be positioned behind the confirmatory statistics, exactly as the existing literature review recommends.

Statistics (primary, confirmatory): generalized linear mixed-effects models, e.g. `decision_outcome ~ attack_type * AI_recommendation_validity + (1|participant) + (1|station)`, which correctly treats the ~20 repeated decisions per participant as non-independent. Effect sizes with confidence intervals, not p-values alone; FDR correction across the many gaze features tested.

Labels: a decision is "risky" only from observable game-log outcomes (chose the compromised kiosk, accepted the fake coupon, lost points) - never inferred from gaze or EEG, which avoids the circular "EEG says risky, therefore risky" argument the review already flags.

Evaluation of any predictive model: participant-level GroupKFold / leave-participants-out cross-validation only; normalization and feature selection fit on training folds alone; report balanced accuracy, macro-F1 and AUROC (risky decisions are the minority class) against a majority-class baseline and a behaviour-only baseline.

What would count as failure, stated up front: if gaze features show no reliable difference between risky and safe decisions after FDR correction, or if a trained model cannot beat the behaviour-only baseline on held-out participants, that is a reportable negative result, not a reason to keep searching for significance.

Implementation roadmap

Quick wins (days, minimal risk)

- Lower `Shadows::textureScalar` and raise `Shadows::disable` in `settings/game-settings-default.cs:30-32`; measure FPS delta before/after with the in-engine overlay.
- Lower `Reflect::refractTexScale` (`game-settings-default.cs:22`) to reduce reflection texture resolution - already applied; needs measuring. (Correction: `Reflect::frameLimitMS` was wrongly flagged as a lever in an earlier pass - its own code comment shows 0 is already the cheapest setting, not uncapped.)
- Set the 2D mirror view to "Empty" if it isn't already, and confirm OpenXR debug mode is off (`lua/ge/extensions/render/openxr.lua:164,174`) - both cheap, zero-risk checks to make before any FPS measurement run.
- Reconcile the two divergent `main.py`/config copies (outer `backend/` vs. deployed `EVChargingGame/backend/`) - they currently differ on `starting_points`, `incident_mode`, and attack probabilities, which is a reproducibility risk for any analysis comparing sessions across dates.

Medium (1-2 weeks)

- Run a proper CPU/GPU/physics profiling pass (RenderDoc / Windows Performance Analyzer) to replace the UNVERIFIED engine-level guesses in Objective 1 with measured numbers.
- EEG decode-and-QC done (76/76 decoded, 67/76 usable - see Objective 2 and Decisions D013/D019). Remaining: ask whoever ran participant 058's session whether its real data exists under a different folder/ID (the current one is a 2-line empty log); for any *new* collection, adapt `collect_data.py` / `OpenMuse` - not `muse.py` - into `receive_ui.py` with session-ID-tagged output on the shared event-marker timeline (Decision D009).
- Build the gaze-only + behaviour-only feature extraction and baseline models (logistic regression, random forest, XGBoost) on the existing 140-session dataset - this doesn't require new data collection or EEG.
- Add the correct/absent/malicious recommendation condition variable to `ai_assistant.py` for the automation-bias study - the VLM kiosk-recommendation system already exists and is directly usable (corrected from an earlier "different

system" characterization, see Decision D017); fix the RAG store's cross-session contamination (D018) before or alongside this.
- Gaze fixation/saccade detection validated on a 3-participant pilot (021, 041, 100) - physiologically plausible results, ready to scale to the remaining 137. AOI mapping (which kiosk/coupon/price someone looked at) needs a team decision first: log kiosk screen-space coordinates going forward, or manually calibrate regions against existing screenshots - it's structurally blocked without one of those, not just unbuilt (Decision D020).

Deep (architectural, next phase)

- Full multimodal pipeline (gaze + EEG + behaviour) once EEG is verifiably collected and synchronized for a meaningful subset of new participants.
- AI-warning integration that actually consumes the trained risk model's output, with the correct/absent/malicious recommendation conditions for the automation-bias sub-study.
- New data collection round incorporating the Objective 3 schema and QC checks above.

Professor presentation summary

A one-page version of this audit's three load-bearing findings, for the next meeting with Prof. Dutt and Ranik Goyal.

1. **The dataset is stronger than the mid-sem report's Table 6 suggests on gaze, and EEG is now a real, decoded, usable dataset for about half the participants, not an open question.** All 140 session folders contain complete Varjo gaze data (~200 Hz, verified by reading the files directly). A separate `backend/Muse/` upload, decoded this pass using the `OpenMuse` library, gives channel-confirmed (TP9/AF7/AF8/TP10 + motion + PPG/fNIRS), sample-rate-confirmed (256.0 Hz exactly) EEG for 76 of 140 participants' raw files; after cross-checking each one's timestamps against its own behavioural session log (not assumed from filenames) and applying a quality threshold, **67 of 140 participants (47.9%) have usable EEG** ready for analysis now. This should replace Table 6's "available, 120+" framing with the real numbers before the next report draft, and the AI pipeline's first pass should still build on gaze+behaviour (full 140-participant coverage) with EEG as a parallel second track on the 67.

2. **The in-sim "AI assistant" already is the kiosk-trust recommendation system the script and report describe - an earlier pass of this audit wrongly called it an unrelated "deepfake-detection chatbot" and that's retracted.** Reading its actual prompts settles it: it screenshots the three kiosks, asks a vision model to flag "deepfake or AI-generated images, unrealistic discounts, phishing prompts, or suspicious branding," and feeds that into an LLM told to recommend the safest kiosk. It's image-based rather than gaze/EEG-based (that part was right), but it is the same system, not a different one. What's genuinely still missing for the automation-bias objective is narrow: a condition variable for a deliberately malicious recommendation - a scoped addition, not a rebuild. One new issue found while verifying this: the AI assistant's memory (RAG store) is shared across all sessions and never cleared automatically between participants - 904 accumulated documents confirmed on disk, spanning dozens of sessions. Worth a direct question to the team about whether `clear_rag_context.py` was run between sessions, since it affects whether participants' AI recommendations can be treated as independent.
3. **FPS optimization is bounded by what's inspectable, and still blocked on a Windows measurement that hasn't happened yet.** BeamNG.tech's renderer and physics engine are closed-source in this install - exact engine-level line numbers, as originally requested, aren't obtainable from this codebase. What is real and actionable: two applied config quick wins (shadow resolution, reflection texture scale), a newly found third one (the desktop mirror view has a developer-labeled "fastest"/"slowest" setting - worth checking it's on fastest), and BeamNG's own native performance-recording tool (`perfProfiler.lua`) and OpenXR debug window, both confirmed to exist by reading the code - neither has been run, because this audit runs on macOS with no access to the Windows/GPU workstation the simulator needs. (One correction: an earlier pass of this audit referenced a profiling command, `gfxFrameOverlay` , that does not exist anywhere in this codebase - retracted.) The 120 FPS target should be revised to a stable 90, matching the XR-4's refresh rate. **This is the one item on this list that only you can unblock** - running the measurement protocol in Objective 1 and sending back the numbers is what turns the rest of Objective 1 from paper into a finding.
Gaze analysis can start now - it's the one pipeline with no open blockers. Fixation/saccade detection was piloted on 3 participants and produces physiologically plausible results (fixation durations 150-250ms, as the literature expects); scaling it to all 140 is a reasonable next step. The one real limitation: mapping gaze to a specific AOI (the coupon vs. the price vs. the kiosk screen) needs either kiosk screen-space coordinates logged going forward or manual region calibration against existing

screenshots - there's no scene geometry or video in the current dataset to ground it automatically, which matches what the literature review itself flagged as the field's hardest open problem.

The literature review and gap analysis (Objective 4) remain the project's strongest asset as-is - no changes recommended beyond verifying the four citations it already flags `[to verify]`.